



Lucy Kao

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Core Skills

COMPUTER VISION

AOI Defect Detection

Anomaly Detection

Image Preprocessing

DEEP LEARNING

YOLO

Model Fine-tuning

PyTorch

BACKEND

Python

FastAPI

MySQL

MongoDB

OTHER

Tableau

Raspberry Pi

Project Leadership

PROJECT MANAGEMENT

Cross-team Coordination

System Rollout

KPI Tracking

OTHER

Corporate Event Planning

Opening Video Production

Education

National Central University

M.S. Information Management
2018 – 2020

Thesis: Collision Avoidance System of Automatic
Vehicle Panorama Dynamic Detection

Shih Hsin University

B.S. Information Management
2012 – 2016

Profile

Senior AI / Software Engineer | 5+ Years in Smart Manufacturing

- Smart Manufacturing & Production-Line AI Deployment:** 5+ years of experience in smart manufacturing and AOI, specializing in pain-point distillation, model fine-tuning, and system deployment — translating production-line needs into stable AI solutions.
- AI Agent Implementation & Application:** Self-directed study of Generative AI, with hands-on experience in LLMs and Tool Use; built automated AI Agent projects, demonstrating rapid adoption of emerging technologies.

Work Experience

Smart Manufacturing Engineer

Oct 2020 – May 2026

AU Optronics (AUO) · Taichung

- Production-Line AI Model Deployment & Yield Optimization:** Fine-tuned models on the existing AOI framework to enable real-time defect detection, accelerating anomaly interception and yield improvement:
 - Used data augmentation and appropriate model selection to address limited defect-sample availability, maintaining model stability.
 - Reconciled gaps between stakeholders' "image features" and "process definitions," refining defect categorization to improve accuracy.
- Requirements Translation & Agile Project Management:** Owned projects end-to-end — from pain-point distillation and technical feasibility assessment to scheduling, system deployment, and KPI tracking.
- Data Interface & Architecture Design:** Led dashboard design, gathered cross-team requirements, and worked with front-end teams to define data specs, building decision-support dashboards.

ERP Systems Developer

Aug 2016 – Sep 2017

Kezhan Information Co. · Taichung

- Customized enterprise ERP modules and forms, integrated cross-system data flows, and optimized interfaces and business logic to ensure stable delivery.

Research Intern — FinTech

Feb 2016 – Jun 2016

Institute for Information Industry · Taipei

- Researched FinTech business models and technology trends; co-authored the industry report "FINTECH66: Selected Cases in Financial Technology Innovation."

Awards

AI CUP 2022 — Orchid Species Identification & Classification

11th Nationally

Competed as part of AU Optronics' team

Led integration of a multi-model ensemble voting mechanism, precisely improving classification performance.

AI CUP 2021 — Automatic Rice Plant Position Labeling on UAV Imagery

5th Nationally

Competed as part of AU Optronics' team

Responsible for high-quality training data annotation and curation, establishing a stable foundation for the model.

Projects

LINE Bot Smart Sales System

Built on FastAPI, integrating Gemini and the Notion API to enable natural-language ordering, CRM, and automated post scheduling.

Receipt OCR Automation System

Combined OCR image recognition with a Notion database to pipe spending data into an automated visualization dashboard.